PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Understanding variations in reported epidemiology of major lower extremity amputation in the UK: A systematic review
AUTHORS	Meffen, Anna; Houghton, John; Nickinson, Andrew; Pepper, Coral; Sayers, Rob; Gray, Laura

VERSION 1 – REVIEW

REVIEWER	Muhammad Zafar Khan Grey's Hospital, General Surgery
REVIEW RETURNED	29-May-2021

GENERAL COMMENTS	Thanks for the opportunity to review your research. I enjoyed reading your article. Unfortunately I do not think it is suitable for publication in its current format. I invite you to review some of my constructive criticism below. 1. Include more details on the burden of diabetes and peripheral arterial disease in your introduction. This will contextualize your paper. 2. Is there a database or a reporting system in the UK for recording amputations. If not perhaps recommend the development of one. 3. Adjust your methodology approach. Consider looking at what the incidence of amputations is among immigrant populations from the subcontinent etc. as well as other specific demographic groups. This may help explain the wide incidence described in your paper. This may help you come to more defined answers in your conclusion. 4. Consider Looking at reamputation rate and any possible factors associated with it. Such as renal failure, Cardiovascular disease etc. 5. Do you have any information on anaemia and transfusion requirements per amputation? This helps stratify the burden of disease of diabetes. 6. Do you have any details on rehabilitation and access to prosthesis.

REVIEWER	Aimee Stewart University of the Witwatersrand School of Social Sciences
REVIEW RETURNED	03-Jun-2021

GENERAL COMMENTS	This is a clear and detailed systematic review. The difficulty of reviewing data that are collected and presented in different formats is detailed. The strengths of the article lie in the care that has been taken in collecting the data that is a thorough review of appropriate articles, analysing the data and interpreting the data. The recommendations made for future reporting of these data are realistic.

REVIEWER	Iván Ascencio-Montiel Instituto Mexicano del Seguro Social
REVIEW RETURNED	16-Jul-2021

GENERAL COMMENTS	The authors present a very well conducted review regarding major lower extremity amputation epidemiology in the United Kingdom. In my opinion, this article is clear and requires only some minor changes: Methods: Given that prevalence-analysis, meta-analysis and meta-regression was not performed due the lack of studies, the last part of the "Data synthesis" section might be removed. Figures: Please state the reason of excluding 3,222 records in the screening section of Figure 1. Instead of a complete date format (dd-mm-yyyy), Figure 2 and Figure 4 X-axis might be shown as year only. Please specify the meaning of blue, red, and green colors in Figure 3.

REVIEWER	Maria Narres Heinrich Heine University Düsseldorf
REVIEW RETURNED	27-Jul-2021

GENERAL COMMENTS	Reviewer comments General comment This is a systematic review aimed to estimate the prevalence/incidence/number of major lower extremity amputations (MLEA) in the UK, to identify sources of routinely collected electronic health data that can be used for surveillance; to assess time trends and regional variation and to identify reasons for variation in reported incidence/prevalence of MLEA. The topic of the article is very interesting and important. However, the methodology in particular as well as the presentation of the results need to be majorly improved. Since the manuscript needs to be completely restructured and profoundly improved, I would opt to reject the manuscript. The following major comments refer to methodological and statistical aspects. Major comments: 1. Introduction Authors wrote, "Peripheral artery disease (PAD) is the leading cause of all major lower extremity amputations (MLEA) with diabetes, smoking, increasing age, hypertension increasing risk.". I disagree with this state because in people with diabetes the diabetic neuropathy is also an important cause of DFU and the following amputation. 2. Methods: Inclusion criteri Authors wrote: "Included reports presented…: a measure of MLEA in adults

	in the general population or in persons with diabetes… ". However, most studies (17 from 24) recorded major amputation in the whole population up to and including all age groups without any age restriction. 3. Methods: Screening and extraction The authors wrote that data were extracted including CI or SE. However, in Table 1 there are no figures with 95% CI or SE. 4. Methods: Data synthesis The authors suggest the following estimates: • Incidence proportion • Incidence rate • Point prevalence • Period prevalence Incidence proportion is defined as Incidence proportion = (Number of new MLEA in the specified time period)/(Total population at risk at the start of the time period) This definition of the term "incidence proportion" is similar to the definition of cumulative incidence. In my opinion, using a new term "incidence proportion" in the context of the incidence of major amputation will lead to confusion in a broad readership of your journal among medical practitioners, epidemiologists and healthcare planners. In reality, it is not always easy to differentiate between the term "incidence rate" where the "population at risk" as person-years was taken into account and other estimates concerning incidence. I disagree with the authors, that in papers by Kennon (9) and Vamos (41) the "incidence proportion" were calculated because there were no clear indications that the population at risk were estimated at the start of the study period. I would assume that in these papers the approximated incidence rates were obtained. In Kennon's paper, the authors mentioned: "…an amputation rate per 1,000 individuals with diabetes per year…" 5. Results 5.1. Search results. Authors wrote: "Twenty-four reports were included, half of which were identified through grey literature searches (4, 7-10, 22, 29-46)". I wonder how it has happened that the authors found the papers from Kennon (36) or Vamos (41), both published in Diabetes Care or Vamos (42), in the grey literature and not in the main search. 5.2. Outcome measures • The authors wrote: "Fourteen (67%) reports provided a calculated statistic,4, 7-10, 29, 31-36, 39-42, the remaining eight provided an absolute number". However 14+8=22. Therefore two papers are lacking. • The authors wrote that the incidence in the general population ranged from 8.8 to 51.1. However, in Table 1, the lowest

	incidence in the general population is 4.9 (Vamos(42)). • Table 1. I think that is very difficult to understand a mix of incidence and the absolute number of amputations. Moreover, the absolute numbers of amputations are presented for one year (43), for two years (44), for three years (22,45,46). All this makes Table 1 too complex, unclear and hard to understand. • In the population with diabetes, the incidence in the study from Holman (8) is 2.51. First, it´s unclear, why this figure was not recalculated as per 100,000 PY. Second, 251 is a figure for all amputations and not for major amputations. The correct incidence is 99 per 100,000 PY. • In the study from Vamos (41), the age-related restriction for people aged 17 years and over are not given. • I don´t understand why the incidence of amputation in people without diabetes from the study by Holman was not shown in Part 1 of Table 1 (general population), although the results in people without diabetes from studies by Vamos (41,42) were here presented. • Additionally, it is unclear why the results in the study by Vamos (41) concerning the incidence in the population with diabetes were not presented in Part 2 (population with diabetes) of Table 1 5.3. Time trend The results regarding the time trend are not presented in the table and are unclearly described in the text. A decreasing time trend was found not only in the study by Kennon but also in the study by Vamos (42). 5.4. Outcome measure calculation differences (page 10 of 61) I don´t understand why the crude incidence in people with diabetes is three times lower compared with standardised incidence: crude rate 87/100,000 and standardised 290/100,000. Amputations in people with diabetes are more common in males and among elderly people. Therefore, I expect that the standardised incidence rate (if the standard general population or non-diabetic population) will be lower than the crude incidence rate. 6. Supplement. Figure 2. The incidence rate for the study by Vamos with approx. 10-15 is wrong presented. The correct incidence is about 100-120 per 100.000 PY.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1:

Comments to the Author:
Thanks for the opportunity to review your research. I enjoyed reading your article.

Unfortunately I do not think it is suitable for publication in its current format.
I invite you to review some of my constructive criticism below.

Comment 1:
Include more details on the burden of diabetes and peripheral arterial disease in your introduction. This will contextualize your paper.

Response:
Thank you for your comment. We have taken this on board and made some additions to the introduction.

Comment 2:
Is there a database or a reporting system in the UK for recording amputations. If not perhaps recommend the development of one.

Response:
There is not a database specific to recording amputations. The secondary care databases HES and NVR should be reporting all MLEA in England and so a database specific to MLEA should not be necessary. Databases used by the included reports are detailed in the section 'Database differences' and Table 2. The effect of database use on epidemiology reporting is discussed in the 'Principle findings' section of the discussion. Specifically, this discusses the drawbacks of the databases with narrow objectives such as the NVR which records only 5 major vascular surgical procedures.

Comment 3:
Adjust your methodology approach. Consider looking at what the incidence of amputations is among immigrant populations from the subcontinent etc. as well as other specific demographic groups. This may help explain the wide incidence described in your paper. This may help you come to more defined answers in your conclusion.

Response:
This is an interesting question, but beyond the scope of this review which had a pre-specified and published protocol. As this review only included reports which studied the whole national population (UK/England/Scotland/Wales or Northern Ireland). All reports included data from every demographic in the population, this would include any immigrant populations residing permanently in the UK who had access to use of the National Health Service. The demographic of the population between the included reports would not have differed in this way and so could not explain the differing incidence proportions between the included reports.

Comment 4:
Consider Looking at reamputation rate and any possible factors associated with it. Such as renal failure, Cardiovascular disease etc.

Response:
The methods applied by the included reports for inclusion/exclusion of multiple procedures per person are discussed under the 'Population difference' section of the results. These methods varied between reports. Investigation of re-amputation rate (and associated risk factors) specifically, whilst an interesting and important topic, were not included in the systematic review objectives or peer reviewed protocol.

Comments 5 & 6:
Do you have any information on anaemia and transfusion requirements per amputation? This helps stratify the burden of disease of diabetes. Do you have any details on rehabilitation and access to prosthesis.

Response:
These were not objectives of the systematic review and not listed in the peer reviewed protocol. The review was also limited by the information provided by the included reports.

Reviewer 2:

Comments to the Author:

This is a clear and detailed systematic review. The difficulty of reviewing data that are collected and presented in different formats is detailed. The strengths of the article lie in the care that has been taken in collecting the data that is a thorough review of appropriate articles, analysing the data and interpreting the data. The recommendations made for future reporting of these data are realistic.

Response:
Thank you very much for your positive comments.

Reviewer 3:

Comments to the Author:
The authors present a very well conducted review regarding major lower extremity amputation epidemiology in the United Kingdom. In my opinion, this article is clear and requires only some minor changes.

Response:
Thank you very much for your positive comments.

Comment 1:
Methods: Given that prevalence-analysis, meta-analysis and meta-regression was not performed due the lack of studies, the last part of the “Data synthesis” section might be removed.

Response:
The PRISMA reporting guidelines require studies to explain how outcome measure synthesis was performed and explain any changes to the originally intended methods stated in the protocol. The Data Synthesis section, along with the explanation as to why the results were narratively synthesised in the Outcome Measures Section in the results, provides this required explanation.

Comment 2:
Figures: Please state the reason of excluding 3,222 records in the screening section of Figure 1.

Instead of a complete date format (dd-mm-yyyy), Figure 2 and Figure 4 X-axis might be shown as year only.

Please specify the meaning of blue, red, and green colors in Figure 3.

Response:
These 3222 records were rejected as they did not fit the inclusion criteria. PRISMA does not require the inclusion of reasons for rejection within the initial screen process.

The date format in Figures 2 and 4 have been amended as recommended.

The colours are to highlight the regions that represent the north, midlands and the south of England that are plotted. This has been clarified in the plot description.

Reviewer 4:

General comment:
This is a systematic review aimed to estimate the prevalence/incidence/number of major lower extremity amputations (MLEA) in the UK, to identify sources of routinely collected electronic health data that can be used for surveillance; to assess time trends and regional variation and to identify reasons for variation in reported incidence/prevalence of MLEA.

The topic of the article is very interesting and important. However, the methodology in particular as well as the presentation of the results need to be majorly improved. Since the manuscript needs to be completely restructured and profoundly improved, I would opt to reject the manuscript.

The following major comments refer to methodological and statistical aspects.

Comment 1:
Introduction: Authors wrote, “Peripheral artery disease (PAD) is the leading cause of all major lower extremity amputations (MLEA) with diabetes, smoking, increasing age, hypertension increasing risk.”. I disagree with this state because in people with diabetes the diabetic neuropathy is also an important cause of DFU and the following amputation.

Response:
Thank you for your comment. We agree that diabetes is also an important cause and included studies that report for the population with diabetes. The introduction has been edited to reflect this.

Comment 2:
Methods: Inclusion criteria Authors wrote: “Included reports presented…: a measure of MLEA in adults in the general population or in persons with diabetes… “. However, most studies (17 from 24) recorded major amputation in the whole population up to and including all age groups without any age restriction.

Response:

Thank you for your comment. We included reports that included adults and excluded reports that only reported for children. We did not exclude those that included both children and adults. We agree that this should could be clearer and have clarified this in the manuscript.

Comment 3:
Methods: Screening and extraction: The authors wrote that data were extracted including CI or SE. However, in Table 1 there are no figures with 95% CI or SE.

Response:
Only two studies gave CI, none SE. This has been clarified in the manuscript and, where available, CI has been added to Table 1.

Comment 4:
Methods: Data synthesis: The authors suggest the following estimates:

• Incidence proportion

• Incidence rate

• Point prevalence

• Period prevalence

Incidence proportion is defined as: Incidence proportion = (Number of new MLEA in the specified time period)/(Total population at risk at the start of the time period)

This definition of the term “incidence proportion” is similar to the definition of cumulative incidence. In my opinion, using a new term “incidence proportion” in the context of the incidence of major amputation will lead to confusion in a broad readership of your journal among medical practitioners, epidemiologists and healthcare planners.

In reality, it is not always easy to differentiate between the term “incidence rate” where the “population at risk” as person-years was taken into account and other estimates concerning incidence. I disagree with the authors, that in papers by Kennon (9) and Vamos (41) the “incidence proportion” were calculated because there were no clear indications that the population at risk were estimated at the start of the study period. I would assume that in these papers the approximated incidence rates were obtained. In Kennon’s paper, the authors mentioned: “…an amputation rate per 1,000 individuals with diabetes per year…”

Response:
Thank you for your comment. We agree that some readers may find the difference between incidence rate and incidence proportion confusing and so have provided referenced definitions within the manuscript for clarification. We decided to use the terms incidence proportion and incidence rate as there are a few commonly used terms for incidence proportion and these two terms were self-explanatory and self-differentiating.

Kennon states:
“The diabetes population was taken from SCI-DC. Incidence of amputation was expressed as number of amputations in people with diabetes per 1,000 individuals with diabetes and per 100,000 general population.” – The denominator counts individuals and not person years, by the referenced definitions this is an incidence proportion.

“This corresponded to a total, major, and minor amputation rate of 2.42, 1.39, and 1.04 per 1,000 individuals with diabetes per year, respectively.” The denominator counts individuals, not person years, by the referenced definition this is an incidence proportion, more specifically, the mean incidence proportion over the study period.

Vamos – Changes states:
“The estimated resident population of England was used to calculate incidence of amputations in individuals without diabetes for each year and incidence rates are expressed per 100,000 people without diabetes.” The denominator is counted as number of individuals in population, not person years, by referenced definition, this is an incidence proportion.

As pointed out, there is no evidence to suggest that the population denominator was not taken as the population at the start of the study period.

Comment 5:
Results: 5.1: Search results: Authors wrote: “Twenty-four reports were included, half of which were identified through grey literature searches (4, 7-10, 22, 29-46)”. I wonder how it has happened that the authors found the papers from Kennon (36) or Vamos (41), both published in Diabetes Care or Vamos (42), in the grey literature and not in the main search.

Response:
The statement quote does not specify which of included articles were identified through grey literature. However, the statement directly after the one quoted does.

“Included grey literature consisted of reports from national healthcare quality improvement programmes, government healthcare department agencies, and government associated health care audits.22, 31-35, 39, 40, 43-46” (This statement has changed slightly in the revised manuscript owing to the addition of reports following the re-run of searches)

Neither of the Vamos reports nor the Kennon report were included in the citations for this statement.

Comment 6:
Results: 5.2: Outcome measures: The authors wrote: “Fourteen (67%) reports provided a calculated statistic,4, 7-10, 29, 31-36, 39-42, the remaining eight provided an absolute number”. However 14+8=22. Therefore two papers are lacking.

Response:
Apologies, this has now been corrected in the manuscript. Thank you for pointing this out.

Comment 7:
Results: 5.2: Outcome measures: The authors wrote that the incidence in the general population ranged from 8.8 to 51.1. However, in Table 1, the lowest incidence in the general population is 4.9 (Vamos(42)).

Response:
The outcome measure ranges for the general population did not include the reports that reported for the population without diabetes. We agree that this could be made clearer and have clarified this in the manuscript.

Comment 8:
Table 1: I think that is very difficult to understand a mix of incidence and the absolute number of amputations. Moreover, the absolute numbers of amputations are presented for one year (43), for two years (44), for three years (22,45,46). All this makes Table 1 too complex, unclear and hard to understand.

Thank you for your comment. Table 1 acts as a summary to highlight not only the outcome values but to highlight the methodological differences between the reports. To this end, all included reports have been presented in the same table. To aid in interpretation of the presented values, the statistic used by each study has also been presented. Not all studies gave a yearly outcome measure for each year of the study period, some gave an outcome measure for the whole of study period. Initially the table had been simplified by only including the outcome value for the most recent year in the study period for each applicable report (as explained in the table footnotes), with studies that provided yearly values discussed in the Time trends section. On request of the reviewer we have presented yearly outcome measure values in Table 1 where available.

Comment 9:
Table 1: In the population with diabetes, the incidence in the study from Holman (8) is 2.51. First, it´s unclear, why this figure was not recalculated as per 100,000 PY. Second, 251 is a figure for all amputations and not for major amputations. The correct incidence is 99 per 100,000 PY.

Response:
We accept that readers may find it convenient for incidence rate values per 10,000 person years to be converted to incidence rate values for 100,000 person years and have changed this in the manuscript. Thank you to the reviewer for pointing out the value error, this has also now been corrected in the manuscript.

Comment 10:
Table 1: In the study from Vamos (41), the age-related restriction for people aged 17 years and over are not given.

Response:
Vamos 2010a (Changes) does not specify any age exclusions for their MLEA cohort. The reference to those aged >17 is in the description of the source of their denominator population data and not their exclusion criteria for the numerator.

Comment 11:
Table 1: I don´t understand why the incidence of amputation in people without diabetes from the study by Holman was not shown in Part 1 of Table 1 (general population), although the results in people without diabetes from studies by Vamos (41,42) were here presented. Additionally, it is unclear why the results in the study by Vamos (41) concerning the incidence in the population with diabetes were not presented in Part 2 (population with diabetes) of Table 1

Response:
Thank you for your comment, we agree and have added these elements to Table 1.

Comment 12:

5.3. Time trends: The results regarding the time trend are not presented in the table and are unclearly described in the text. A decreasing time trend was found not only in the study by Kennon but also in the study by Vamos (42).

Response:
The trends by Vamos are acknowledged in the manuscript in the sentence below however the reference is incorrect and should be Vamos 2010b (Trends) report as the reviewer suggests. This has now been corrected.

“Although results are mixed, with some reports finding non-significant decreasing trends, there is some evidence of a decrease in MLEA in England over time for both the general population and for those with diabetes.42”

Additionally, yearly outcome measures have now presented in Table 1.

Comment 13:
5.4. Outcome measure calculation differences (page 10 of 61) I don´t understand why the crude incidence in people with diabetes is three times lower compared with standardised incidence: crude rate 87/100,000 and standardised 290/100,000. Amputations in people with diabetes are more common in males and among elderly people. Therefore, I expect that the standardised incidence rate (if the standard general population or non-diabetic population) will be lower than the crude incidence rate.

Response:
We agree with the reviewer that this is not an expected result. These values were stated in the report and not calculated by the reviewers. Unfortunately the report did not provide any detailed methods that would allow the reviewers to unpick and understand the difference in crude and standardised incidence proportions presented.

Comment 14:
6. Supplement. Figure 2. The incidence rate for the study by Vamos with approx. 10-15 is wrong presented. The correct incidence is about 100-120 per 100.000 PY.

Response:
Apologies and thank you for your comment. This has now been corrected.

VERSION 2 – REVIEW

REVIEWER	Muhammad Zafar Khan Grey's Hospital, General Surgery
REVIEW RETURNED	07-Sep-2021

GENERAL COMMENTS	Thanks for the opportunity to review this paper after major revision. I think a good effort has been made to address some of my previous comments. Areas that could not be addressed or are beyond the context of the protocol have been stated such.